

Semantic-Aware Hierarchical 3D Gaussian Representation for Autonomous Driving Simulation

Abstract—3D Gaussian Splatting (3DGS) has recently emerged as an efficient paradigm for high-fidelity urban scene reconstruction in autonomous driving simulation. However, existing methods rely on globally uniform representations, leading to severe redundancy and inefficient resource allocation in large-scale scenes. Moreover, limited viewpoints and occlusions result in insufficient supervision, causing blurry textures and degraded multi-view consistency. To address these challenges, we propose a semantic-aware hierarchical 3D Gaussian representation that jointly optimizes anchor distribution and level-of-detail (LOD) across different semantic regions. Specifically, we introduce a semantic-guided anchor allocation strategy to adaptively control Gaussian density, along with a category-aware LOD mechanism for efficient multi-scale rendering. To further alleviate supervision scarcity, we incorporate diffusion-based pseudo-view augmentation to enrich training signals under occlusions. In addition, a multi-view consistency loss is designed to enforce geometric and photometric alignment across views. Building upon these designs, we establish a complete pipeline from 3D reconstruction to simulation data synthesis. Extensive experiments demonstrate that our method reduces GPU memory usage while maintaining a competitive training cost.

I. INTRODUCTION

High-fidelity scene reconstruction and simulation are fundamental to the development, testing, and validation of autonomous driving systems. In practical closed-loop simulation, a reconstructed scene should not only provide photorealistic rendering, but also support efficient novel-view synthesis, object-level manipulation, and scenario editing under different camera trajectories and traffic configurations. Recently, 3D Gaussian Splatting (3DGS) [1] has attracted increasing attention because it combines explicit point-based geometry with differentiable rasterization, achieving real-time rendering with high visual fidelity.

However, directly applying existing 3DGS-based representations to large-scale driving scenes remains challenge. First, urban scenes are highly heterogeneous. Roads, sky regions, distant buildings, and vegetation usually contain large smooth or repetitive areas, while vehicles, pedestrians, lane markings, traffic signs, and semantic boundaries require much finer geometric and appearance details. Existing methods often allocate Gaussian primitives in a globally uniform or purely photometric manner, which easily causes redundant primitives in simple background regions and insufficient modeling capacity around safety-critical foreground objects and semantic boundaries. This imbalance increases memory and rendering cost, and weakens the deployability of 3DGS in autonomous driving simulation.

Second, driving data are captured from vehicle-mounted cameras with constrained trajectories and limited baselines.

Many regions are observed only from narrow viewpoints or are frequently occluded by dynamic actors. As a result, the reconstruction is weakly supervised in disoccluded areas, side surfaces of vehicles, distant structures, and regions behind moving objects. This insufficient supervision often leads to blurred textures, floaters, unstable geometry, and poor multi-view consistency. These artifacts are especially harmful for simulation, where downstream perception and planning modules are sensitive to geometric alignment, object boundaries, and view-consistent appearance.

To address these issues, we propose a semantic-aware hierarchical Gaussian representation for efficient and high-quality reconstruction of autonomous driving scenes. Our key observation is that different semantic categories should not be modeled with the same Gaussian budget or the same rendering granularity. Dynamic and foreground actors require explicit, editable Gaussian primitives to preserve fine geometry and support object-level simulation. In contrast, large static backgrounds can be represented more compactly by structured neural Gaussian anchors, from which local view-adaptive Gaussians are generated only when needed. Based on this dual-mode design, we further introduce semantic-guided anchor allocation and category-aware level-of-detail control, allowing the model to allocate more capacity to complex or safety-critical regions while keeping smooth background areas compact.

Moreover, to alleviate the sparse-view and occlusion problem, we incorporate diffusion-based pseudo-view generation as an additional supervisory signal. The generated pseudo-views enrich observations for under-constrained regions, while the proposed multi-view consistency loss regularizes geometry and appearance across real and pseudo views. In this way, the framework jointly improves reconstruction fidelity, rendering efficiency, and view consistency, making it suitable for photorealistic simulation and flexible asset insertion in autonomous driving scenarios. The main contributions of this work are summarized as follows:

- We introduce neural Gaussian anchors for background modeling and further propose a semantic-aware hierarchical Gaussian representation. By jointly optimizing anchor distribution and category-aware level of detail, the proposed method adaptively allocates modeling capacity to different scene components and improves the efficiency-quality trade-off in large-scale driving scenes.
- We propose a generative-prior-enhanced weakly supervised reconstruction strategy. Diffusion-based pseudo-views provide additional supervision for occluded and sparsely observed regions, while a multi-view consis-

tency loss improves geometric and photometric alignment across views.

- We build a complete autonomous-driving-oriented reconstruction and simulation pipeline that supports high-quality 3DGS reconstruction, photorealistic novel-view rendering, and flexible scene editing. Extensive experiments on public datasets demonstrate superior reconstruction and synthesis quality in most scenarios.

II. RELATED WORK

Recent 3D Gaussian Splatting (3DGS) methods have shown strong potential for autonomous driving scene reconstruction due to their high-fidelity and real-time rendering capability. StreetGS [2] decomposes dynamic urban scenes into static backgrounds and foreground vehicles for efficient composition and editing. HUGS [3] jointly models geometry, appearance, semantics, and motion, while S³Gaussian [4] explores self-supervised street-scene decomposition. However, these methods mainly focus on dynamic-static decomposition, pose modeling, or object-level manipulation, and still lack fine-grained adaptive allocation of Gaussian primitives across semantic categories.

Semantic information has also been incorporated into Gaussian representations. Gaussian Grouping [5], FMGS [6], and Semantic Gaussians [7] enhance object grouping, scene understanding, or open-vocabulary querying by embedding semantic features into Gaussians. Nevertheless, semantics in these methods is primarily used for segmentation or editing, rather than for guiding primitive distribution, rendering granularity, or level-of-detail control. Structured Gaussian representations further improve scalability. Scaffold-GS [8] introduces neural anchors to generate view-adaptive Gaussians, Octree-GS [9] organizes Gaussians with an LOD octree, and CityGaussian [10] adopts divide-and-conquer training with level-of-detail strategies for large-scale scenes. These methods reduce redundancy, but are generally category-agnostic and do not fully exploit the semantic heterogeneity of autonomous driving scenes. In contrast, our framework combines semantic guidance with hierarchical Gaussian modeling: foreground actors are represented by explicit editable Gaussians, while large static backgrounds are modeled with compact neural anchors, enabling category-aware allocation and level-of-detail control for efficient high-quality reconstruction.

III. METHOD

A. Overview

Our goal is to reconstruct a high-fidelity and efficient 3D Gaussian representation for autonomous driving scenes from multi-view RGB images and LiDAR point clouds. As illustrated in Fig. 1, we address two key challenges: (1) inefficient global representations in large-scale scenes, and (2) insufficient supervision caused by occlusions and limited viewpoints.

To this end, we propose a **semantic-aware hierarchical Gaussian representation** that unifies spatial distribution and level-of-detail (LOD) optimization. The overall framework

consists of three components: 1) semantic-aware hierarchical representation for efficient modeling, 2) diffusion-based pseudo-view augmentation for enhanced supervision, and 3) multi-view consistency optimization for robust reconstruction.

B. Semantic-Aware Hierarchical Gaussian Representation

1) *Semantic-Guided Anchor Allocation*: Instead of using a globally uniform Gaussian distribution, we allocate Gaussian primitives based on semantic structure to better match scene complexity. Specifically, we partition the static background into semantic groups:

$$G = \{G_{\text{ground}}, G_{\text{tree}}, G_{\text{car}}, G_{\text{other}}\}. \quad (1)$$

Dynamic objects are excluded using bounding boxes to obtain a clean background point cloud. For each semantic group g , we assign a category-specific density coefficient α_g , which controls the number of Gaussian anchors:

$$N_g = \max(\lfloor N_{\text{base}} \cdot \alpha_g \rfloor, 1). \quad (2)$$

Intuitively, complex and high-frequency regions (e.g., trees) are allocated more anchors, while smooth regions (e.g., ground) are sparsely represented, improving efficiency without sacrificing detail. We then apply Farthest Point Sampling (FPS) to generate anchors:

$$A_g = \text{FPS}(P_g, N_g), \quad (3)$$

which ensures spatial coverage and avoids clustering compared to uniform or voxel-based initialization. The final anchor set is:

$$A = \bigcup_g A_g. \quad (4)$$

2) *Category-Aware Level-of-Detail (LOD)*: To further improve efficiency, we introduce a semantic-aware LOD mechanism that adaptively controls rendering granularity. For each anchor a_i , we compute a normalized distance ratio:

$$\bar{d}_i = \min\left(\frac{d_i}{d_{\text{max}}}, 1\right), \quad (5)$$

and define a semantic-aware effective distance:

$$\hat{d}_i = \beta_{l_i} \cdot \bar{d}_i, \quad (6)$$

where β_{l_i} is a category-specific multiplier. This design assigns larger effective distances to less critical regions (e.g., ground), encouraging early pruning of fine details while preserving important structures. The maximum LOD level is then computed as:

$$k_i^* = \left\lfloor (1 - \hat{d}_i) \cdot k_{\text{max}} \right\rfloor. \quad (7)$$

During rendering, only anchors satisfying $k_i \leq k_i^*$ are activated, enabling coarse-to-fine adaptive rendering.

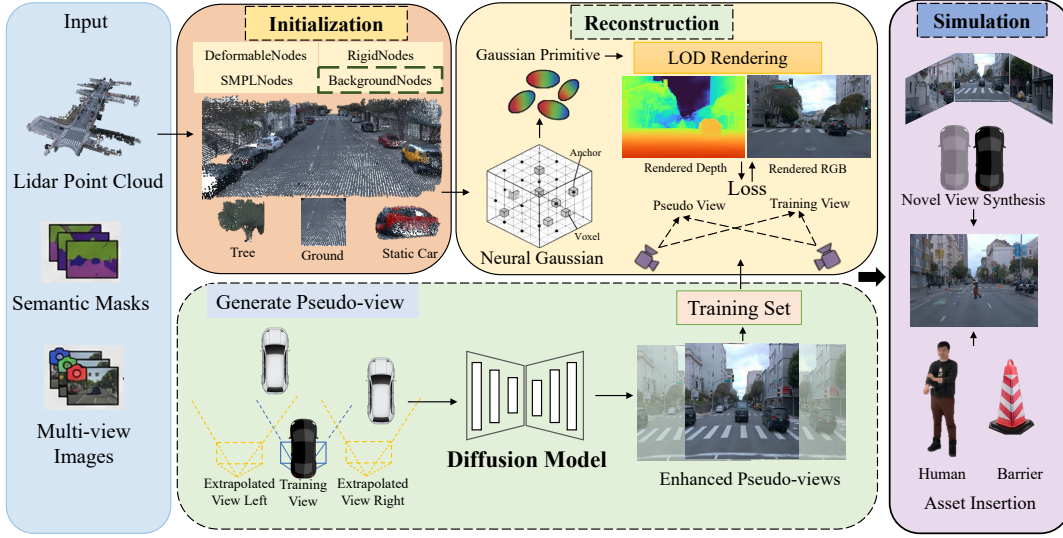


Fig. 1. Overview of our method. Given multi-view RGB images, semantic masks, and LiDAR point clouds, we initialize category-aware background anchors to construct neural 3D Gaussians and apply semantic-aware LOD control for efficient reconstruction. Diffusion-generated pseudo-views provide additional supervision, enabling high-quality 3DGS reconstruction, photorealistic novel-view rendering, and flexible asset insertion for autonomous driving simulation.

3) *Dynamic Anchor Optimization*: To maintain a compact yet expressive representation, we perform dynamic anchor refinement during training. When splitting anchors, child anchors inherit semantic labels and progressively increase detail levels:

$$k_{\text{child}} = \min(k_p + 1, k_{\text{max}}). \quad (8)$$

Conversely, anchors with low contribution are pruned. Importantly, semantic labels and LOD levels are updated synchronously to ensure consistency across attributes. This mechanism enables scalable optimization while preserving structural coherence.

C. Diffusion-Based Pseudo-View Augmentation

To alleviate supervision scarcity caused by occlusions and sparse viewpoints, we introduce a generative prior-guided pseudo-view augmentation strategy. Specifically, we first generate novel viewpoints by interpolating along camera trajectories. These views are then refined using a diffusion model Difix3D+ [11] to produce geometrically and semantically consistent images. The generated pseudo-views are progressively incorporated into the training set. This strategy effectively enriches supervision signals in under-observed regions and reduces overfitting to limited viewpoints, leading to improved reconstruction completeness and robustness.

D. Multi-View Consistency Optimization

To enforce consistency across views, we design a multi-view consistency loss that jointly constrains geometry and appearance. The overall training objective is:

$$\begin{aligned} \mathcal{L} = & (1 - \lambda_r)\mathcal{L}_1 + \lambda_r\mathcal{L}_{\text{SSIM}} + \lambda_d\mathcal{L}_{\text{depth}} \\ & + \lambda_o\mathcal{L}_{\text{opacity}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{mv}} \end{aligned} \quad (9)$$

The multi-view consistency loss is defined as:

$$\begin{aligned} \mathcal{L}_{\text{mv}} = & \lambda_{\text{geo}} \frac{1}{|M|} \sum_{p \in M} w_p \|p - \hat{p}\|_2^2 \\ & + \lambda_{\text{ncc}} \frac{1}{|M|} \sum_{p \in M} w_p (1 - \text{NCC}(I_1(p), I_2(p))) \end{aligned} \quad (10)$$

where the first term enforces geometric consistency via reprojection error, and the second term enforces photometric consistency using normalized cross-correlation. The adaptive weight w_p suppresses outliers based on reprojection uncertainty.

E. Simulation-Oriented Scene Construction

Building upon the reconstructed Gaussian representation, we further support flexible asset insertion for autonomous driving simulation. Pre-optimized Gaussian assets (e.g., pedestrians, vehicles) can be seamlessly integrated into the scene while maintaining geometric and photometric consistency as shown in Fig. 2. This enables the generation of diverse and photorealistic simulation data, facilitating downstream perception and planning tasks.

IV. EXPERIMENTS

A. Datasets and Evaluation Metrics

We evaluate our method on three widely used autonomous driving datasets: Waymo Open Dataset [13], KITTI [14], and nuScenes [15], covering diverse urban environments and sensor configurations. Waymo is used as the primary benchmark for quantitative and qualitative evaluation, while KITTI and nuScenes are adopted to validate cross-dataset generalization.

Following prior works, we evaluate rendering quality using Peak Signal-to-Noise Ratio (PSNR) [16], Structural Similarity Index (SSIM) [17], and Learned Perceptual Image

TABLE I
QUANTITATIVE COMPARISON ON WAYMO, KITTI AND nuSCENES DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Model	Waymo			KITTI			nuScenes		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
PVG	29.42	0.86	0.22	26.22	0.89	0.06	28.29	0.82	0.25
OmniRe	31.16	0.90	0.18	28.18	0.91	0.08	28.38	0.85	0.22
Ours	32.52	0.92	0.15	29.77	0.92	0.05	28.67	0.84	0.24



Fig. 2. Asset Insertion Examples. We provide four categories of assets (car, obstacle, human [12], animal) for autonomous driving scene augmentation, which can be inserted into 3D reconstructed scenes to simulate real-world driving environments.

Patch Similarity (LPIPS) [18]. Higher PSNR and SSIM indicate better reconstruction fidelity, while lower LPIPS reflects improved perceptual similarity. In addition, we report GPU memory consumption and training time to assess computational efficiency.

B. Implementation Details

Our model is trained for 30k iterations using multi-view RGB images and LiDAR point clouds. The training process is divided into two stages: an initial reconstruction stage and a pseudo-view augmented stage. Diffusion-based pseudo-views are progressively incorporated after 20k iterations to enhance supervision. We set the maximum LOD level $k_{\max} = 4$ and the maximum observation distance $d_{\max} = 100$. Category-specific density coefficients α_g and distance multipliers β_{l_i} are empirically determined based on scene complexity. All experiments are conducted on a single GPU A800.

C. Baselines

We conducted comprehensive comparative experiments between our proposed method and several state-of-the-art 3D Gaussian-based methods on the Waymo dataset. To verify cross-scenario generalization, we also evaluate on KITTI and nuScenes. The baseline methods are selected to cover the major technical paradigms in dynamic street-scene rendering. StreetGS [2] and PVG [19] represent foundational works that pioneer the application of 3D Gaussian Splatting to dynamic outdoor scenes. BezierGS [20] introduces explicit trajectory-aware modeling via Bézier curves for dynamic agents. OmniRe [21] serves as a comprehensive strong baseline with a unified framework for diverse dynamic object categories. Together, these methods span the full spectrum from early

TABLE II
QUANTITATIVE COMPARISON ON WAYMO DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Model	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
StreetGS	31.96	0.92	0.14
BezierGS	25.23	0.84	0.28
PVG	29.42	0.86	0.22
OmniRe	31.16	0.90	0.18
Ours	32.52	0.92	0.15

approaches to current state-of-the-art, enabling a thorough and fair evaluation of our method.

D. Comparison with State-of-the-Art

Quantitative Results. As shown in Table I and Table II, our method achieves the best PSNR on all evaluated datasets and obtains competitive SSIM and LPIPS results. On Waymo, our approach reaches a PSNR of 32.52 and an SSIM of 0.92, outperforming OmniRe in both metrics. Although StreetGS obtains a slightly lower LPIPS on Waymo, our method provides a better overall balance between reconstruction fidelity, perceptual quality, and memory efficiency.

Qualitative Results. As illustrated in Fig. 3, our method generates sharper object boundaries, clearer textures, and more consistent geometry than baseline methods. In occluded and complex regions, our approach effectively reduces blurring and distortion. On KITTI and nuScenes, our method remains competitive or superior, showing strong generalization ability under different data distributions and sensor setups as shown in Fig. 4.

E. Efficiency Analysis

We analyze efficiency in terms of GPU memory usage and training time in Table III. Compared with OmniRe, our method reduces GPU memory consumption while achieving higher reconstruction quality. The additional pseudo-view supervision slightly increases training time, but improves texture sharpness and multi-view consistency in occluded regions. Overall, the proposed semantic-aware LOD strategy provides a favorable trade-off between reconstruction quality and memory efficiency for large-scale scenes.

F. Visualization

We visualize the training process in Fig. 5. Compared with OmniRe, our method converges faster and produces clearer structures at earlier iterations, benefiting from better



Fig. 3. Qualitative results on Waymo dataset

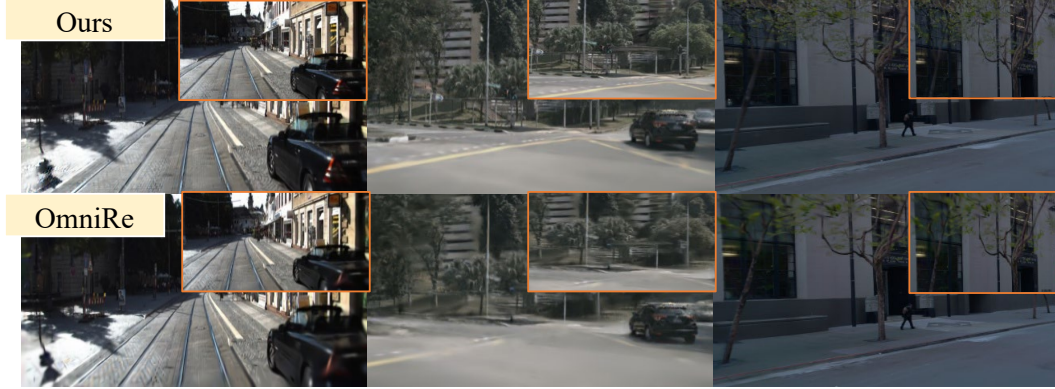


Fig. 4. Qualitative comparison of rendering results between our method and OmniRe on KITTI (left), nuScenes (middle), and Waymo (right) datasets. The zoomed-in regions highlight that our method produces sharper object details, clearer texture, and more accurate geometric structure compared to OmniRe, especially in challenging street scenes with complex lighting and dynamic objects.



Fig. 5. Visualization of rendering results during training. We compare our method and OmniRe at different training steps. As training progresses, our method achieves clearer scene details, sharper object edges, and more accurate geometric structure compared to OmniRe, demonstrating faster convergence and better rendering quality.

representation design and enhanced supervision. To enrich simulation scenarios, we leverage SAM3D [22] and GART [23] to obtain high-quality Gaussian assets including vehicles, pedestrians, and obstacles. As shown in Fig. 6, our framework supports seamless asset insertion with consistent geometry, appearance, and occlusion relations, enabling flexible and photorealistic autonomous driving simulation data generation.

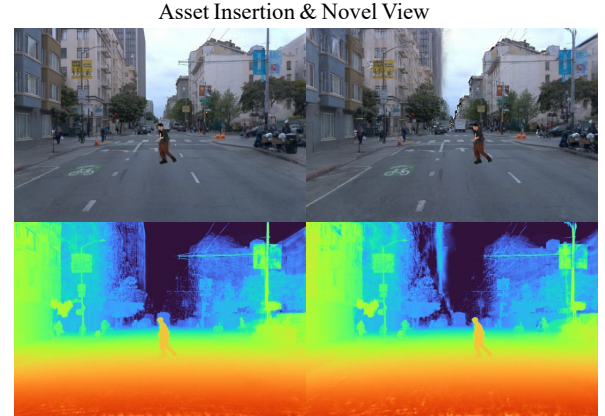


Fig. 6. Qualitative demonstration of asset insertion and novel view rendering. Top row: Rendered RGB images of a street scene with an inserted human asset, viewed from original view and novel view. Bottom row: Corresponding depth maps, which verify the accurate geometric consistency of the inserted asset with the 3D scene structure across novel views. Our method enables photorealistic asset insertion with correct occlusion, lighting, and perspective.

G. Ablation Study

To validate each core component, we perform ablation experiments on the Waymo dataset (Table III). Removing any module leads to clear performance drops, with increased blurriness and structural inconsistency. Ablating semantic-

TABLE III

QUANTITATIVE COMPARISON OF PERFORMANCE AND EFFICIENCY, AND ABLATION STUDY ON KEY COMPONENTS.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	GPU Memory (GB)	Training Time
OmniRe	29.73	0.88	0.21	18.19	42min
Ours (Full)	31.72	0.92	0.14	14.94	48min
w/o category-aware LOD	30.24	0.89	0.17	18.34	46min
w/o Diffusion Augmentation	30.81	0.90	0.18	14.63	31min
w/o Multi-view Loss	30.12	0.88	0.19	14.82	46min

aware anchor allocation results in fragmented scene structures. Disabling category-aware LOD increases redundancy and degrades fine details. Removing diffusion-based pseudo-view augmentation leads to unstable textures and incomplete reconstruction in occluded areas. Removing multi-view consistency loss causes geometric misalignment and view inconsistency. Our full model achieves the best overall performance.

V. CONCLUSION

This paper proposes a semantic-aware anchored 3D Gaussian splatting framework. The method combines semantic-driven adaptive anchor assignment with category-aware level-of-detail control to balance reconstruction quality and computational efficiency, and introduces generative prior enhancement with multi-view consistency constraints to improve weakly supervised and occluded regions. Experimental results show that our method outperforms mainstream baselines across multiple metrics while maintaining efficient rendering, demonstrating strong potential for high-quality reconstruction and realistic simulation data generation in autonomous driving scenarios.

REFERENCES

- [1] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3d gaussian splatting for real-time radiance field rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, July 2023. [Online]. Available: <https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/>
- [2] Y. Yan, H. Lin, C. Zhou, W. Wang, H. Sun, K. Zhan, X. Lang, X. Zhou, and S. Peng, “Street gaussians for modeling dynamic urban scenes,” *arXiv preprint arXiv:2401.01339*, 2024.
- [3] H. Zhou, J. Shao, L. Xu, D. Bai, W. Qiu, B. Liu, Y. Wang, A. Geiger, and Y. Liao, “Hugs: Holistic urban 3d scene understanding via gaussian splatting,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 21 336–21 345.
- [4] N. Huang, X. Wei, W. Zheng, P. An, M. Lu, W. Zhan, M. Tomizuka, K. Keutzer, and S. Zhang, “S³gaussian: Self-supervised street gaussians for autonomous driving,” *arXiv preprint arXiv:2405.20323*, 2024.
- [5] M. Ye, M. Danelljan, F. Yu, and L. Ke, “Gaussian grouping: Segment and edit anything in 3d scenes,” in *European conference on computer vision*. Springer, 2024, pp. 162–179.
- [6] X. Zuo, P. Samangouei, Y. Zhou, Y. Di, and M. Li, “Fmgs: Foundation model embedded 3d gaussian splatting for holistic 3d scene understanding,” *International Journal of Computer Vision*, vol. 133, no. 2, pp. 611–627, 2025.
- [7] J. Guo, X. Ma, Y. Fan, H. Liu, and Q. Li, “Semantic gaussians: Open-vocabulary scene understanding with 3d gaussian splatting,” *IEEE Transactions on Circuits and Systems for Video Technology*, 2026.
- [8] T. Lu, M. Yu, L. Xu, Y. Xiangli, L. Wang, D. Lin, and B. Dai, “Scaffold-gs: Structured 3d gaussians for view-adaptive rendering,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 20 654–20 664.
- [9] K. Ren, L. Jiang, T. Lu, M. Yu, L. Xu, Z. Ni, and B. Dai, “Octree-gs: Towards consistent real-time rendering with lod-structured 3d gaussians,” *arXiv preprint arXiv:2403.17898*, 2024.
- [10] Y. Liu, C. Luo, L. Fan, N. Wang, J. Peng, and Z. Zhang, “Citygaussian: Real-time high-quality large-scale scene rendering with gaussians,” in *European Conference on Computer Vision*. Springer, 2024, pp. 265–282.
- [11] J. Z. Wu, Y. Zhang, H. Turki, X. Ren, J. Gao, M. Z. Shou, S. Fidler, Z. Gojcic, and H. Ling, “Difix3d+: Improving 3d reconstructions with single-step diffusion models,” in *CVPR*, 2025, pp. 26 024–26 035.
- [12] W. Jiang, K. M. Yi, G. Samei, O. Tuzel, and A. Ranjan, “Neuman: Neural human radiance field from a single video,” in *European Conference on Computer Vision*. Springer, 2022, pp. 402–418.
- [13] P. Sun, H. Kretschmar, X. Dotiwalla, A. Chouard, V. Patnaik, P. Tsui, J. Guo, Y. Zhou, Y. Chai, B. Caine *et al.*, “Scalability in perception for autonomous driving: Waymo open dataset,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 2446–2454.
- [14] A. Geiger, P. Lenz, and R. Urtasun, “Are we ready for autonomous driving? the kitti vision benchmark suite,” in *2012 IEEE conference on computer vision and pattern recognition*. IEEE, 2012, pp. 3354–3361.
- [15] H. Caesar, J. Kabzan, K. S. Tan, W. K. Fong, E. Wolff, A. Lang, L. Fletcher, O. Beijbom, and S. Omari, “nuplan: A closed-loop ml-based planning benchmark for autonomous vehicles,” *arXiv preprint arXiv:2106.11810*, 2021.
- [16] U. Sara, M. Akter, M. S. Uddin *et al.*, “Image quality assessment through fsim, ssim, mse and psnr—a comparative study,” *Journal of Computer and Communications*, vol. 7, no. 3, pp. 8–18, 2019.
- [17] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, “Image quality assessment: from error visibility to structural similarity,” *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.
- [18] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, “The unreasonable effectiveness of deep features as a perceptual metric,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 586–595.
- [19] Y. Chen, C. Gu, J. Jiang, X. Zhu, and L. Zhang, “Periodic vibration gaussian: Dynamic urban scene reconstruction and real-time rendering,” *International Journal of Computer Vision*, vol. 134, no. 3, p. 83, 2026.
- [20] Z. Ma, J. Jiang, Y. Chen, and L. Zhang, “Bézierngs: Dynamic urban scene reconstruction with bézier curve gaussian splatting,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 25 519–25 528.
- [21] Z. Chen, J. Yang, J. Huang, R. De Lutio, J. M. Esturo, B. Ivanovic, O. Litany, Z. Gojcic, S. Fidler, M. Pavone *et al.*, “OmniRe: Omni urban scene reconstruction,” *arXiv preprint arXiv:2408.16760*, 2024.
- [22] M. S. D. Team, “Sam 3d: 3dfy anything in images,” *arXiv preprint arXiv:2511.16624*, 2025.
- [23] J. Lei, Y. Wang, G. Pavlakos, L. Liu, and K. Daniilidis, “Gart: Gaussian articulated template models,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 19 876–19 887.